**Sheet 2** Mon. 12 Oct. 2026 Determinant of two vectors; collinearity criterion

**1.** Warm-up · Flash Calculate $2 \times (-3) - 4 \times 5$ and $(-1) \times 6 - 3 \times (-2)$.

**2.** Core · Method Calculate $\det(\vec{u},\vec{v})$ for $\vec{u}\begin{pmatrix}2\\-3\end{pmatrix}$ and $\vec{v}\begin{pmatrix}-4\\6\end{pmatrix}$. What can we conclude?

**3.** Core + · True/false True or false? Justify. The vectors $\vec{u}\begin{pmatrix}1\\2\end{pmatrix}$ and $\vec{v}\begin{pmatrix}3\\5\end{pmatrix}$ are collinear.

**4.** Back · Construction steps Draw a circle with centre $O$ and two perpendicular diameters $[AC]$ and $[BD]$. What type of $ABCD$ ? Justify.

**5.** Running thread · Flash Procedure: choose a number; multiply it by 4; add 5. a. What result for 3 ? b. Which expression in terms of the starting number $x$ ?

**Sheet 3** Tue. 13 Oct. 2026 Collinearity and parallelism: the determinant method

**1.** Warm-up · Flash Calculate the coordinates of $\overrightarrow{AB}$ for $A(1;-2)$ and $B(4;3)$.

**2.** Core · Method The points $A(1;2)$, $B(3;3)$ and $C(7;5)$ collinear?

**3.** Core + · Method For which value of $k$ the vectors $\vec{u}\begin{pmatrix}k\\3\end{pmatrix}$ and $\vec{v}\begin{pmatrix}4\\6\end{pmatrix}$ collinear?

**4.** Back · Flash Calculate 20% of 350.

**5.** Running thread · Translate A rectangle has length $x+3$ and of width $x$ (in cm). Express its perimeter as a function of $x$ and simplify it.

**Sheet 4** Thu. 15 Oct. 2026 Problem solving: vector, coordinate or geometric method

**1.** Warm-up · Flash Complete: $ABCD$ is a parallelogram $\Leftrightarrow \overrightarrow{AB} = \ldots$

**2.** Core · Method $A(0;0)$, $B(4;1)$, $C(5;4)$. Determine $D$ so that $ABCD$ is a parallelogram.

**3.** Core + · Write up $A(0;0)$, $B(3;1)$, $C(-1;3)$. Prove that $ABC$ is right-angled and isosceles at $A$.

**4.** Back · Flash Write 450,000 in scientific notation.

**5.** Running thread · Flash How many faces, vertices and edges does a cube have?

## Lines in the plane

**Sheet 5** Mon. 2 Nov. 2026 Direction vector of a line

**1.** Warm-up · Flash Give a vector collinear with $\vec{u}\begin{pmatrix}2\\-4\end{pmatrix}$ with simpler integer coordinates.

**2.** Core · Method Give a direction vector of $(AB)$ for $A(1;2)$ and $B(4;8)$.

**3.** Core + · MCQ A direction vector of the line through $A(0;1)$ and $B(2;-3)$ is:

**A.** $\begin{pmatrix}1\\-2\end{pmatrix}$ **B.** $\begin{pmatrix}1\\-4\end{pmatrix}$ **C.** $\begin{pmatrix}-4\\-2\end{pmatrix}$ **D.** $\begin{pmatrix}2\\4\end{pmatrix}$

**4.** Back · Flash Marks: 8, 12, 12, 15, 18. Calculate the mean, the median and the range.

**5.** Running thread · Property $ABC$ is right-angled at $A$ and $I$ is the midpoint of $[BC]$. What can be said about $IA$, $IB$ and $IC$ ? Why?

**Sheet 6** Tue. 3 Nov. 2026 Cartesian equation of a line

**1.** Warm-up · Flash Calculate $3x - 2y + 1$ for $x = 1$ and $y = -2$.

**2.** Core · Method The line $d$ has equation $2x + 3y - 6 = 0$. Give a direction vector and a point of $d$.

**3.** Core + · True/false True or false? Justify. The lines $2x + 3y - 6 = 0$ and $4x + 6y + 1 = 0$ are parallel.

**4.** Back · Table Mode of transport of 30 pupils.

| Transport | Bus | Bicycle | Walking | Car |
|---|---|---|---|---|
| Count | 12 | 5 | 8 | 5 |

Give the share of pupils coming by bus and the angle of the "bicycle" sector in a pie chart.

**5.** Running thread · Flash On a number line, $A$ has x-coordinate $-2$ and $B$ with x-coordinate 3. Give $AB$ and the x-coordinate of the midpoint of $[AB]$.

**Sheet 7** Thu. 5 Nov. 2026 Reduced equation and slope of a non-vertical line

**1.** Warm-up · Flash Isolate $y$ in 2 $x + y - 5 = 0$.

**2.** Core · Method Give the slope and y-intercept of the line 3 $x - 2y + 4 = 0$.

**3.** Core + · MCQ The slope of the line $(AB)$ with $A(1;2)$ and $B(5;0)$ is:

**A.** 2 **B.** $\frac{1}{2}$ **C.** $-2$ **D.** $-\frac{1}{2}$

**4.** Back · Flash Calculate $\frac{3}{4} + \frac{5}{6}$.

**5.** Running thread · Problem Two gears have 12 and 18 teeth. After how many turns of each do they return together to their starting position? After how many turns of each

**Sheet 8** Mon. 9 Nov. 2026 Draw a line given by an equation

**1.** Warm-up · Flash For $y = -2x + 3$, calculate $y$ when $x = 0$, then when $x = 2$.

**2.** Core · Method Give two points for drawing the line 2 $x + y - 4 = 0$.

**3.** Core + · Flash What is the line with equation $x = 3$ ? Give a direction vector. Does it have a slope?

**4.** Back · Flash Compare $\frac{3}{7}$ and $\frac{4}{9}$

**5.** Running thread · Flash Solve in $\mathbb{R}$ : $5x - 7 = 2x + 8$.

**Sheet 9** Tue. 10 Nov. 2026 Determine an equation of a line: two points, point and direction vector, point and slope

**1.** Warm-up · Flash Slope of the line through $(1;3)$ and $(3;7)$ ?

**2.** Core · Method Reduced equation of the line with slope 2 passing through $A(1;3)$.

**3.** Core + · Method Line through $A(1;2)$ with direction vector $\vec{u}\begin{pmatrix}2\\-1\end{pmatrix}$ : Cartesian equation by the determinant of $\overrightarrow{AM}$ and $\vec{u}$.

**4.** Back · Flash Simplify $2^{5} \times 2^{-3}$.

**5.** Running thread · Table → formula An affine function $f$ satisfies: $f(0) = -1, f(1) = 1, f(2) = 3$. Give $f(x)$.

**Sheet 10** Thu. 12 Nov. 2026 Lines and affine functions: linking the representations

**1.** Warm-up · Flash $f(x) = -3x + 4$. Calculate $f(2)$ and the preimage of 10.

**2.** Core · Method Determine the affine function $f$ such that $f(1) = 1$ and $f(3) = 5$.

**3.** Core + · True/false True or false? Justify. Every line in the plane is the graph of an affine function.

**4.** Back · Flash A fair 6-sided die is rolled. Probability of getting a multiple of 3?

**5.** Running thread · Flash $ABC$ is right-angled at $A$ with $AB = 6$ cm and $AC = 8$ cm. Calculate $BC$.